

Statement of Contribution

1. General Role in the Project

I helped in collecting previous research papers for the literature review. I also studied, cleaned, and analyzed the Boston dataset and provided insightful analysis on pollution patterns. Additionally, I thoroughly understood the full scope of the project, including my colleagues' parts, which enabled me to contribute effectively to writing both the Introduction and Methodology sections of the IEEE paper.

2. Technical Contribution

- Collected and reviewed previous research papers relevant to urban air quality analysis.
- Loaded and cleaned the Boston_PM_NO2.csv dataset using pandas.
- Converted time formats and applied missing value treatment techniques.
- Used H3 hexagonal encoding to spatially aggregate the Boston data.
- Conducted exploratory data analysis including histograms for PM2.5, NO2, temperature, and humidity.
- Implemented bootstrapping to estimate mean NO2 distribution and assess statistical reliability.
- Performed KMeans clustering on H3-level data to detect spatial pollution patterns.
- Built a multivariate linear regression model to predict NO2 based on PM2.5, temperature, and humidity.
- Exported final analysis outputs to boston_summary_stats.csv and boston_aggregated_h3.csv.
- Supported team member by validating outputs from other cities and ensuring code consistency across the project.

"I have attached screenshots of my code to clearly demonstrate my hands-on work in reference section "

3. IEEE Paper Contribution

I contributed to writing and editing the following key sections of the IEEE paper:

- **Introduction:** I wrote the full introduction, including the background on urban air quality, the impact of PM2.5, and the motivation behind comparing New York, Boston, and Chicago.

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Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

- **Methodology:** I contributed to the data collection and preprocessing parts of the methodology, specifically describing how the Boston dataset was handled, cleaned, and spatially encoded using H3 indexing.
- **Boston Analysis Support:** I conducted the full technical analysis for the Boston dataset and provided detailed outputs (EDA, regression, clustering, and bootstrapping), which were later integrated into the results section by another team member.

“Please refer to the pink-highlighted sections in the attached IEEE paper in reference section”

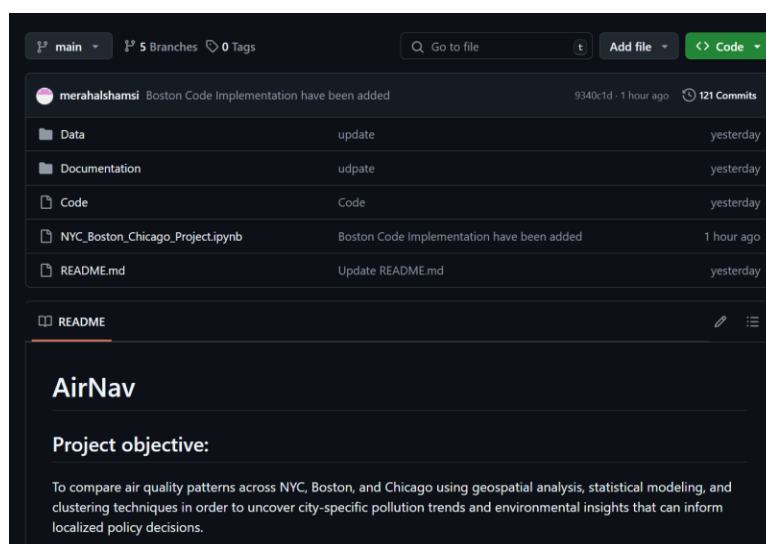
4. GitHub Pull Requests (PRs) & Repo Structure & Organization

I contributed directly to the main shared Jupyter Notebook file titled NYC_Boston_Chicago_Project.ipynb, where I implemented and documented the full Boston analysis pipeline. This includes data loading, cleaning, H3 spatial encoding, clustering, regression modeling, and visualizations.

Instead of opening a separate pull request, our team used a **shared-edit workflow**, where each member committed their own part directly to the main branch.

Additionally, I participated in organizing and maintaining the GitHub repository to ensure smooth collaboration across the team:

- I contributed to the Data/ folder by uploading the Boston dataset.
- I helped build the Documentation/ folder by adding the algorithm explanations, formulas, and references to previous research papers we used.



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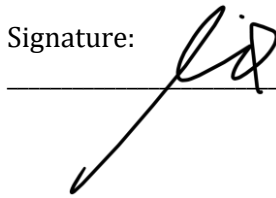
6. Peer Review & Team Involvement

Throughout the project, I stayed actively involved with my teammates by reviewing their work, giving feedback, and making sure everything stayed consistent and aligned with our overall goals. I regularly joined team discussions, supported others in refining their parts, and helped ensure that our work was cohesive and presented in a professional and structured way. I also contributed to organizing and maintaining the shared project files, making collaboration smoother for everyone

Declaration

I confirm this document truthfully represents my contribution to the group project.

Signature:

A handwritten signature in black ink, consisting of stylized cursive letters, is written over a horizontal line.

Reference:

A Comparative Spatial Analysis of Urban Air Quality: New York City, Boston, and Chicago

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Abstract—Urban air quality is a significant health and environmental challenges of the 21st century. This research uses advanced geospatial techniques and statistical analysis to conduct comparative studies of air quality in three major US metropolitan cities: New York City, Boston, and Chicago. It employs advanced geospatial techniques and statistical analysis to examine variations in particulate matter (PM_{2.5}, PM₁₀) and nitrogen dioxide (NO₂) and how these are related to weather patterns. We integrate disparate environmental data with hexagonal space indexing, machine learning cluster algorithms, and multi-variable regression models to capture meaningful patterns in air pollution and city attributes. Results indicate that there are noteworthy city-wide variations in pollution, with Chicago showing high values of PM_{2.5}, New York City showing high PM₁₀ variations, and Boston showing comparatively low pollutant means. These results emphasize the requirement of city-specific air quality control and present an empirical foundation in support of evidence-based environmental policy design.

Index Terms—air quality, PM_{2.5}, urban pollution, spatial analysis, machine learning, GIS, environmental health

I. INTRODUCTION

Air pollution is one of the most serious challenges that all governments worldwide are confronted with, with far-reaching impacts on public health, quality of life, and environmental sustainability. Fine particulate matter (PM_{2.5}) – those particles with aerodynamic dimensions of less than or equal to 2.5 micrometers – are most health-hazardous because such particles are capable of reaching deep into the lungs and entering the bloodstream. These extremely minute particles, finer than hair by approximately 30-fold, are from several sources like exhausts from vehicles, industrial activities, combustion, and natural environments like wildfires and dust storms.

Health effects of PM_{2.5} have been well established, ranging from temporary airway irritation to chronic health conditions like the exacerbation of asthma, decreased lung function, cardiovascular disease, and increased mortality in vulnerable groups. Environmental protection agency regulations in most places regulate air quality in accordance with PM_{2.5} concentration levels in the following ways:

- Concentrations below 12 $\mu\text{g}/\text{m}^3$ are generally linked with good air quality and minimal health risks.
- Values between 12–35 $\mu\text{g}/\text{m}^3$ represent moderate risk and may affect sensitive persons.
- Concentrations greater than 35 $\mu\text{g}/\text{m}^3$ are unhealthy and pose population health risks.

Despite the recognized importance of air quality monitoring and control, comparative studies in urban areas are fraught with methodological challenges. Cities are heterogeneous in fundamental configuration on physical layout, population density, industrial composition, transportation patterns, and weather patterns. There are also variations in methods of data collection, siting strategies of sensors, and frequencies of monitoring that pose challenges in making cross-city comparisons. The variations necessitate that there should be one standardized analysis method to allow meaningful comparative analysis of urban air quality.

Our research responds to these challenges by developing a Geographic Information Systems (GIS) strategy that normalizes the evaluation of air pollution prevalence in New York City, Boston, and Chicago. While utilizing H3 hexagonal spatial indexing – a geospatial system that divides geographic space into regular hexagonal cells – we define a uniform space that negates structural dissimilarity in urban environments. This enables us to compare air pollution prevalence proportionately, analyze air pollution hotspots, and compare weather impacts in differing metropolitan environments.

Whereas traditional air quality monitoring often resorts to simple averaging or static mapping, we leverage the use of K-means clustering algorithms to identify the discrete regimes of pollution in every urban area. This unsupervised machine learning method captures hidden patterns in the space that can be overlooked by traditional approaches. We further explore the intricate dependency between environment factors and amounts of pollution with the assistance of multivariate regression modeling and observe the influence of temperature and humidity on amounts of PM_{2.5} in different urban environ-

Their findings on the varying impact of urban form metrics depending on city development stages support our intent to compare multiple cities with different urban structures.

Safarov et al. [10] applied advanced GIS-based interpolation and structural equation modeling in Pavlodar and Kazakhstan to quantify spatial pollution patterns and their relationship with environmental variables. Their use of mobile sensors and geospatial modeling illustrated the effectiveness of spatially-resolved air quality analysis. This is a method we replicate using mobility and pollution data from different cities.

Collectively, these research works offer key methodologies and frameworks to synthesize and analyze geospatial big data in urban mobility and environmental settings. They note the challenges and methods involved in handling large-scale data, maintaining real-time speeds, and extracting actionable intelligence from intricate urban data sets. These works are the foundation of how we compare the patterns of mobility across cities and measure the consequence on air quality, and ultimately, what policy recommendations are put forth to enhance urban sustainability.

III. METHODOLOGY

A. Data Collection

1) *Data Source and Features:* To conduct a comparative spatial analysis of New York City, Boston, and Chicago air quality, we downloaded publicly available sensor-based environmental data for each city. The data include major air pollutants (PM2.5, NO, PM10, CO, and O) and contextual environmental variables (temperature, humidity), geospatial coordinates (latitude, longitude), and time stamps.

New York City data was collected from five files with data on PM2.5, PM10, temperature, humidity, and sensor locations. Boston's dataset included PM2.5 and NO levels along with temperature and humidity. Chicago's data, sampled from five components too included the most general pollutant coverage of PM2.5, CO, calibrated NO, and O readings, along with meteorological and device-related variables such as battery and signal strength.

2) *Data Collection Limitations and Challenges:* This study is related to the lack of overlap between pollutant data in the three cities. PM10 data was available only in New York City, and O and CO were available only in Chicago, precluding the possibility of consistent multi-variable comparison across all the datasets. This lack of overlap limited the generalizability of some of the analyses and required city-specific modeling. Moreover, although all datasets included time-stamped measurements, differences in sampling frequency and sensor calibration may have introduced heterogeneity in data quality and temporal accuracy. A further challenge was heterogeneity in the amount of data across different cities, with some sites having a much greater quantity of sensor readings than others, potentially influencing the results. Sparse metadata on sensor types and deployment criteria reduced transparency in data reliability.

To surmount these limitations, we used city-specific models, implemented H3 hexagonal indexing to normalize spatial

comparisons, and relied on statistical validation (e.g., RMSE, bootstrapping) to enhance result reliability.

B. Data Processing and Engineering

1) *Data Cleansing and Outlier Handling:* We developed a comprehensive preprocessing pipeline for time and weather data in raw datasets. Early issues included various time stamp formats, missing values for key variables (e.g., PM2.5, temperature, humidity), and extreme outliers from sensor malfunctions. Time stamps were centralized so cities have a shared time format. Missing values in the dataset were eliminated or imputed via forward-fill and mean imputation.

To ensure validity of data, measurements outside domain-specific ranges were filtered out. These included PM2.5 measurements outside 0–500 $\mu\text{g}/\text{m}^3$, temperatures outside -30°C to 60°C , and humidity outside 0% to 100%. These preprocessing measures were undertaken to ensure the reliability of downstream spatial aggregation, clustering, and regression modeling, and to ensure equitable comparison of air quality trends in New York City, Boston, and Chicago.

2) *Spatial Aggregation Using H3 Indexing:* One of the key challenges in this study is that cities differ significantly in layout, size, sensor density, and climate, making it difficult to compare air quality data fairly across regions. To address this, we applied H3 hexagonal spatial indexing, a technique developed by Uber that transforms geographic coordinates into uniform, area-based spatial units. By encoding all sensor locations into hexagonal cells of equal size, we ensured that data from New York City, Boston, and Chicago could be aggregated and compared within a consistent spatial framework.

We chose H3 resolution 9, representing zones of about 0.105 km^2 , ideal for neighbourhood environmental analysis. As shown in Table I, H3 provides a range of resolutions, each aligned with a specific spatial use case—from city-level overviews to street-level detail. Using this method, each hex cell aggregates air quality and environmental measurements (e.g., PM2.5, temperature, humidity), allowing us to conduct city-wide comparisons without bias from differing urban structures. Algorithm 1 details converting raw data into structured spatial grids for clustering and regression analysis.

TABLE I: H3 Resolution and Spatial Coverage

Resolution	Average Hexagon Area (km^2)	Typical Use Case
7	~ 5.16	City-wide overviews
8	~ 0.74	District-level analysis
9	~ 0.105	Neighborhood-level studies
10	~ 0.015	Street-level granularity
11	~ 0.0021	Building-level assessments

3) *Aggregation and Feature Engineering:* After assigning each observation to its corresponding hexagonal zone using the H3 index at resolution 9, we performed spatial aggregation to generate structured input features for modelling. For each

ments.

We account for the variations in environmental data through the use of robust validation techniques like bootstrapping and Root Mean Square Error (RMSE) testing to ensure maximum statistical reliability and accuracy of predictions. These techniques strengthen the robustness of findings and ensure greater confidence in the relationships modeled between weather conditions and pollutant concentration.

With intuitive visualizations like heatmaps, scatterplots, and 3D regression surfaces, we turn complex environmental data into actionable insights that inform policymakers, researchers, and public health officials. Our strategy is to answer some of the most fundamental questions regarding urban air quality:

- 1) What are the differences in air pollution concentration patterns in major cities?
- 2) Which cities have the most $PM_{2.5}$ air pollution burdens?
- 3) Which are the strongest indicators of weather conditions that are related to air quality deterioration?
- 4) Can commutative capacity of air quality measures among urban environments be improved by analytical methods based on GIS?

The framework that has been created in this study has data-driven and scalable procedures to carry out integrated air quality assessment in urban settings. Increased accuracy in cross-city comparison and identification of determinants of pollutant attributes is made possible with the proposed approach, allowing evidence-based decision support to frame environmental policy, resource planning, and intervention measures towards improving urban air quality and disease prevention.

II. RELATED WORK

The combination of mobility patterns and environmental conditions, in this case air quality, is an important aspect of grasping how city dynamics create and contribute to climatic changes. With increasing urban expansion, understanding how mobility patterns are affecting the environmental parameters like air pollution becomes relevant. This project extends this line of reasoning by comparing mobility patterns of different cities and determining how these impact air quality. A number of important studies form the basis for this work, providing methodologies and findings that are immediately relevant to what we are doing.

Al Jawarneh et al. [1] introduced a scalable method to combine heterogeneous mobility and air pollution data with QoS guarantees. They have a highly effective model that is capable of efficiently handling big geospatial data to support joint analysis on multi-faceted urban datasets. This directly influences how we're handling mobility and environmental data in such a manner that guarantees the robustness of our analysis, especially while comparing air quality between cities.

Further, Al Jawarneh et al. [2] were concerned with approximate query processing in smart cities, utilizing spatial data streams to process mobility and environmental conditions in real-time. This highlights the importance of effective processing methods capable of withstanding the constant stream of data produced by urban systems. This is in line with our work

since real-time analysis of mobility and air quality will play a crucial role in getting actionable insights that will inform policy measures.

Another pertinent work by Al Jawarneh et al. [3] is on the use of mobility and environmental information in climate change analytics. This work investigates how the mobility of urban areas may be analyzed together with environmental parameters in order to offer a more holistic view of how cities are affected by climate change. The work of theirs emphasizes the need to integrate several sources of data to investigate the intricate association between urban mobility and environmental conditions—an approach that will be followed in our project.

Al Jawarneh et al. [4] also made a valuable contribution by emphasizing the simplification of sophisticated geospatial data towards efficient analytics. The research conducted by them on polygon simplification of georeferenced big data will be useful in the handling of big datasets like those with mobility and air quality data in cities. Simplifying the spatial data will assist us in enhancing the efficiency and scalability of analysis back in our own work.

In addition, the research by Ghaffarpasand et al. [5] explores how urban mobility affects air pollution in Kampala, a sub-Saharan city. The authors integrated mobility data from GPS and air quality monitoring to model transportation patterns against the concentration of pollutants. The study employs statistical modeling in order to estimate how mobility in the urban environment contributes to pollutant concentration. This research is very relevant to our project because it is insightful regarding how mobility patterns influence air quality, especially in cities with heterogeneous urban dynamics.

Bandeira and colleagues [6] analyzed the influence of land use on mobility patterns and emissions in a Portuguese city of medium size. Traffic data collection, land use categorization, and air pollutant monitoring were used in their work to determine how the variety of land use categories impacts mobility and emissions. This approach is very relevant to our research because knowing how land use drives mobility patterns will allow us to compare metropolitan environments in different cities and compare how these environments contribute to air quality.

Furthermore, Austin et al. [7] applied k-means clustering to group U.S. air monitoring sites based on long-term $PM_{2.5}$ composition. They showed how source types and geography affect air pollution profiles. Their study highlighted the value of clustering multivariate pollution data. This informs city-level clustering methods for mobility and air quality patterns.

Ross et al. [8] suggested a hybrid land use regression and temporal adjustment model for estimating exposure to $PM_{2.5}$ and NO_2 across NYC. Their detailed spatial modeling using health outcome data highlights the importance of high-resolution environmental exposure estimation. This parallels our city-level geospatial aggregation using H3 indices.

Liang and Gong [9] conducted a large-scale study across 626 Chinese cities. They analyzed how different urban forms (compact vs. dispersed) influence long-term $PM_{2.5}$ trends.

▼ Boston Datasets

STEP 1: Load Boston Data

```
from google.colab import files
uploaded = files.upload()
```

```
boston_df = pd.read_csv("Boston_PM_N02.csv")
print("Boston columns:", boston_df.columns)
print("Shape:", boston_df.shape)
boston_df.head()
```



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cell to enable.

Saving Boston_PM_N02.csv to Boston_PM_N02.csv

```
Boston columns: Index(['SensorID', 'time', 'latitude', 'longitude', 'bin0', 'bin1', 'bin10',
'bin11', 'bin12', 'bin13', 'bin14', 'bin15', 'bin16', 'bin17', 'bin18',
'bin19', 'bin2', 'bin20', 'bin21', 'bin22', 'bin23', 'bin3', 'bin4',
'bin5', 'bin6', 'bin7', 'bin8', 'bin9', 'humidity', 'temperature',
'pm25', 'no2'],
dtype='object')
Shape: (122768, 32)
```

	SensorID	time	latitude	longitude	bin0	bin1	bin10	bin11	bin12	bin13	...	bin4	bin5	bin6	bin7	bin8	bin
0	Boston_CS01A	1644587952	42.406059	-71.116470	85	8	0	1	0	0	...	0	0	0	0	0	
1	Boston_CS01A	1644587957	42.406059	-71.116470	87	5	0	0	0	0	...	1	0	0	0	0	
2	Boston_CS01A	1644587962	42.406059	-71.116478	89	10	0	0	0	0	...	1	1	0	0	0	
3	Boston_CS01A	1644587967	42.406059	-71.116486	113	10	0	0	0	0	...	0	1	0	0	0	
4	Boston_CS01A	1644587972	42.406063	-71.116501	73	8	0	0	0	0	...	0	1	0	0	0	

STEP 2: Time conversion

```
boston_df["datetime"] = pd.to_datetime(boston_df["time"], unit="s")
```

STEP 3: Apply H3 Geohash

```
boston_df["h3"] = boston_df.apply(lambda row: h3.geo_to_h3(row["latitude"], row["longitude"], 9), axis=1)
```

STEP 4: EDA - Histograms for Boston

```
plt.figure(figsize=(16, 6))
```

PM2.5

```
plt.subplot(2, 2, 1)
sns.histplot(boston_df["pm25"].dropna(), bins=50, kde=True)
plt.title("Boston - PM2.5 Distribution")
plt.grid(True)
```

NO2

```
plt.subplot(2, 2, 2)
sns.histplot(boston_df["no2"].dropna(), bins=50, kde=True)
plt.title("Boston - NO2 Distribution")
plt.grid(True)
```

Temperature

```
plt.subplot(2, 2, 3)
sns.histplot(boston_df["temperature"].dropna(), bins=50, kde=True)
plt.title("Boston - Temperature")
plt.grid(True)
```

Humidity

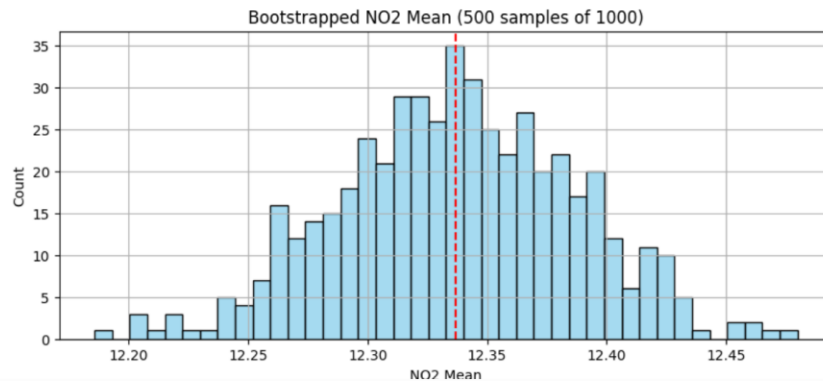
```
plt.subplot(2, 2, 4)
sns.histplot(boston_df["humidity"].dropna(), bins=50, kde=True)
plt.title("Boston - Humidity")
plt.grid(True)
```

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NYC_Boston_Chicago.ipynb - Colab

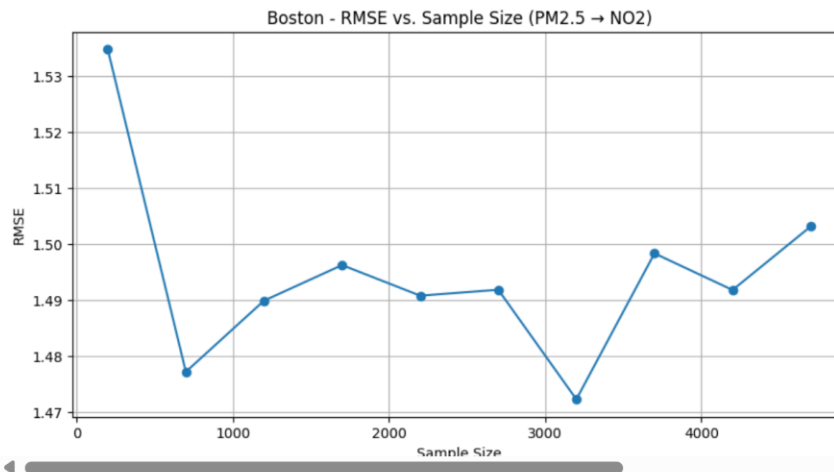
```
plt.figure(figsize=(10, 4))
sns.histplot(boot_means_no2, bins=40, color='skyblue')
plt.axvline(np.mean(boot_means_no2), color='red', linestyle='--')
plt.title("Bootstrapped NO2 Mean (500 samples of 1000)")
plt.xlabel("NO2 Mean")
plt.grid(True)
plt.show()
```



```
# STEP 7: RMSE vs Sample Size
samples = list(range(200, min(5000, len(boston_df)), 500))
rmse_boston = []
```

```
for size in samples:
    sample = boston_df.sample(n=size)
    X = sample[["pm25"]].fillna(0)
    y = sample["no2"].fillna(0)
    model = LinearRegression().fit(X, y)
    preds = model.predict(X)
    rmse = np.sqrt(mean_squared_error(y, preds))
    rmse_boston.append(rmse)

plt.figure(figsize=(10, 5))
plt.plot(samples, rmse_boston, marker='o')
plt.title("Boston - RMSE vs. Sample Size (PM2.5 → NO2)")
plt.xlabel("Sample Size")
plt.ylabel("RMSE")
plt.grid(True)
plt.show()
```



Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

4/30/25, 10:59 PM

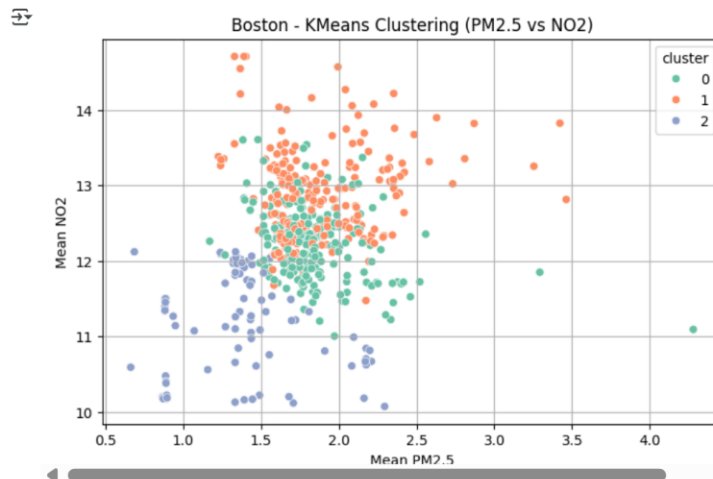
NYC_Boston_Chicago.ipynb - Colab

```
# STEP 8: Clustering - Aggregate by H3
agg_boston = boston_df.groupby("h3").agg({
    "pm25": "mean",
    "no2": "mean",
    "temperature": "mean",
    "humidity": "mean"
}).reset_index()
agg_boston.columns = ["h3", "pm25_mean", "no2_mean", "temp_mean", "humidity_mean"]

# Clustering
features = ["pm25_mean", "no2_mean", "temp_mean", "humidity_mean"]
X = agg_boston[features].dropna().reset_index(drop=True)
X_scaled = StandardScaler().fit_transform(X)

kmeans = KMeans(n_clusters=3, random_state=42)
agg_clustered_boston = agg_boston.dropna(subset=features).reset_index(drop=True)
agg_clustered_boston["cluster"] = kmeans.fit_predict(X_scaled)

# Plot clusters
plt.figure(figsize=(8, 5))
sns.scatterplot(data=agg_clustered_boston, x="pm25_mean", y="no2_mean", hue="cluster", palette="Set2")
plt.title("Boston - KMeans Clustering (PM2.5 vs NO2)")
plt.xlabel("Mean PM2.5")
plt.ylabel("Mean NO2")
plt.grid(True)
plt.show()
```



```
# STEP 9: Regression
df_boston = boston_df[["no2", "pm25", "temperature", "humidity"]].dropna()
X_reg_boston = sm.add_constant(df_boston[["pm25", "temperature", "humidity"]])
y_reg_boston = df_boston["no2"]
```

```
model_boston = sm.OLS(y_reg_boston, X_reg_boston).fit()
print(model_boston.summary())
```

```
OLS Regression Results
=====
Dep. Variable:      no2      R-squared:      0.393
Model:              OLS      Adj. R-squared: 0.393
Method:             Least Squares      F-statistic: 2.651e+04
Date:               Wed, 30 Apr 2025      Prob (F-statistic): 0.00
Time:               16:31:51      Log-Likelihood: -1.9359e+05
No. Observations:   122768      AIC: 3.872e+05
Df Residuals:       122764      BIC: 3.872e+05
Df Model:            3
Covariance Type:    nonrobust
=====
               coef      std err      t      P>|t|      [0.025      0.975]
-----
const          10.6383      0.013    819.520      0.000      10.613      10.664
pm25            0.1289      0.004     29.050      0.000      0.120      0.138
temperature     -0.0826      0.001    -134.748      0.000     -0.084     -0.081
humidity         0.0587      0.000    255.431      0.000      0.058      0.059
=====
```

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4/30/25, 10:59 PM

NYC_Boston_Chicago.ipynb - Colab

```

Omnibus:            3463.218    Durbin-Watson:            0.028
Prob(Omnibus):      0.000    Jarque-Bera (JB):      3274.734
Skew:               0.357    Prob(JB):              0.00
Kurtosis:           2.639    Cond. No.              168.
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```

# STEP 10: Save Summary Stats + Clustering
summary_boston = df_boston.describe()[["pm25", "no2"]].T[["mean", "std", "min", "max"]]
summary_boston["city"] = "Boston"
agg_clustered_boston["city"] = "Boston"

summary_boston.to_csv("boston_summary_stats.csv")
agg_clustered_boston.to_csv("boston_aggregated_h3.csv", index=False)

print("Boston analysis complete and saved.")

```

 Boston analysis complete and saved.

▼ Chicago Datasets

```

# STEP 1: Load and merge all Chicago eclipse data parts
import pandas as pd

from google.colab import files
uploaded = files.upload()

files = [
    "chicago_eclipse_data_part_1.csv",
    "chicago_eclipse_data_part_2.csv",
    "chicago_eclipse_data_part_3.csv",
    "chicago_eclipse_data_part_4.csv",
    "chicago_eclipse_data_part_5.csv"
]

chicago_df = pd.concat([pd.read_csv(f) for f in files], ignore_index=True)

print("Chicago shape:", chicago_df.shape)
print("Columns:", chicago_df.columns)
chicago_df.head()

```



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```

Saving chicago_eclipse_data_part_1.csv to chicago_eclipse_data_part_1.csv
Saving chicago_eclipse_data_part_2.csv to chicago_eclipse_data_part_2.csv
Saving chicago_eclipse_data_part_3.csv to chicago_eclipse_data_part_3.csv
Saving chicago_eclipse_data_part_4.csv to chicago_eclipse_data_part_4.csv
Saving chicago_eclipse_data_part_5.csv to chicago_eclipse_data_part_5.csv
Chicago shape: (647655, 16)
Columns: Index(['City', 'DeviceId', 'LocationName', 'Latitude', 'Longitude',
               'ReadingDateTimeUTC', 'PM25', 'CalibratedPM25', 'CalibratedO3',
               'CalibratedNO2', 'CO', 'Temperature', 'Humidity', 'BatteryLevel',
               'PercentBattery', 'CellSignal'],
              dtype='object')

```

	City	DeviceId	LocationName	Latitude	Longitude	ReadingDateTimeUTC	PM25	CalibratedPM25	CalibratedO3	CalibratedN
0	Chicago	2002	State & Garfield (SB)	41.794921	-87.625857	2021-06-20 00:03:00	5.561094	NaN	NaN	Ni
1	Chicago	2002	State & Garfield (SB)	41.794921	-87.625857	2021-06-20 00:08:10	6.633914	NaN	NaN	Ni
2	Chicago	2002	State & Garfield (SB)	41.794921	-87.625857	2021-06-20 00:13:20	4.068707	NaN	NaN	Ni

```

# STEP 2: Rename and prepare
chicago_df.rename(columns={
    "Latitude": "latitude",
    "Longitude": "longitude",
    "ReadingDateTimeUTC": "datetime",
    "PM25": "pm25",
    "Temperature": "temperature",
    "Humidity": "humidity"
})

```

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Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago